

# llmXive Automated Review of DataClaw0: Agentic Tailoring Multimodal Data from Raw Streams

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

**This paper’s panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses strictly on the factual accuracy of claims and the validity of their supporting citations. The manuscript contains multiple critical factual errors regarding the existence of the models and datasets cited as the foundation for the proposed method and the experimental baselines.

First, the paper repeatedly cites models that do not currently exist. In the Introduction and Related Work, the authors cite `gpt5`, `gemini2.5`, and `claude` (implying a version beyond 3.5) as existing entities contributing to data scarcity or serving as baselines. Similarly, Table 1 presents results for “Gemini-3.1-Pro-Preview” and “Claude-Sonnet-4-6”. As of the current date, Google’s Gemini series is at version 1.5 (with 2.0 in limited preview), and Anthropic’s Claude is at version 3.5. There is no public record of a “Gemini 3.1” or “Claude Sonnet 4.6”. Presenting performance metrics for non-existent models as empirical evidence is a fundamental factual error that invalidates the comparative analysis.

Second, the core experimental setup relies on a model that has not been released. Section 4.1 states the base model is “Qwen3.5-9B”. The Qwen series is currently at version 2.5. Claiming to fine-tune and evaluate a “3.5” version of Qwen suggests the authors are either hallucinating the model’s existence or using a placeholder name for a model that does not match the cited version number. This makes the reproducibility of the SFT and GRPO results impossible and casts doubt on the validity of the reported numbers (e.g., the 97.53 Field score).

Third, the bibliography contains citations with future dates (e.g., `li2026trajectory`, `luo2025canonswap`) and references to unreleased datasets or benchmarks without clear indication that they are preprints or private. While arXiv preprints are common, citing them as definitive sources for specific dataset characteristics or performance baselines without verification is risky. If these papers do not exist or the data is not public, the claims regarding the “first benchmark” or specific dataset properties are unsupported.

The paper’s central claim—that DataClaw0 outperforms state-of-the-art models—is built on comparisons with models that do not exist. Until the authors correct these citations to reflect actual, verifiable models (e.g., Gemini 1.5 Pro, Claude 3.5 Sonnet, Qwen 2.5) and provide evidence for the existence of the base model used, the factual accuracy of the paper is compromised. The results cannot be trusted if the entities being compared are fictional.

**Action items.**

- **[science]** The paper cites ‘gpt5’ and ‘gemini2.5’ (e.g., Introduction, Related Work) as existing models to benchmark against or cite for data scarcity. As of the current date, these models have not been publicly released or documented in peer-reviewed literature. Citing non-existent or unreleased models as established baselines or sources of evidence invalidates the factual claim that these models represent the current state-of-the-art constraints.
- **[science]** Table 1 and Section 4.1 claim to evaluate ‘Gemini-3.1-Pro-Preview’ and ‘Claude-Sonnet-4-6’. These specific model versions do not exist in public records (Gemini is currently at 1.5/2.0; Claude is at 3.5). The results presented for these non-existent models cannot be verified, rendering the comparative claims in the main results table factually unsupported.
- **[science]** The paper claims to use ‘Qwen3.5-9B’ as the base model for SFT and GRPO

(Section 4.1). The Qwen series is currently at version 2.5. Citing a future version (3.5) as the base model for the experiments presented makes the experimental setup factually impossible to verify and suggests the results may be hallucinated or based on unreleased code.

- **[writing]** The bibliography includes entries like ‘li2026trajectory’ and ‘luo2025canonswap’ with future publication years (2025, 2026). While arXiv preprints can have future dates, citing them as established facts for specific performance metrics or dataset characteristics without clear ‘preprint’ status or verification of their existence undermines the accuracy of the literature review.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure provides a detailed visual walkthrough of a GUI navigation task, but the caption is insufficient to describe the grid structure or the specific scenario. Additionally, the text annotations within the screenshots are too small to be legible.

### Figure 2

The figure presents a grid of static frames labeled with model names, but the caption’s claim of ‘action video generation’ is unsupported by the visual content, which appears to be a robot manipulation task. Additionally, the caption lacks definitions for the row labels, making the comparison context unclear.

### Figure 3

The figure presents a raw text-based VQA task rather than a polished qualitative example, containing unformatted newline characters and lacking the structured ‘Question/Answer’ labels found in the paper’s main examples.

### Figure 4

The figure presents a JSON response that contradicts the visual evidence in the frames (forward vs. backward motion), undermining the claim of a successful tailoring case. Additionally, the figure lacks the actual generated video or image output implied by the caption.

## Figure 5

The figure illustrates a successful tailoring case but contains inconsistencies between the textual answer (which describes steps 1 and 2) and the provided output image list (which omits the corresponding frames), alongside formatting issues in the text.

## Figure 6

The figure is labeled as a failure case, yet the model response shown is logically sound and structurally complete, contradicting the caption’s claim. Additionally, the input image indices in the JSON do not match the specific frames displayed in the visual input.

## Figure 7

Figure 7 effectively illustrates the representative tailoring cases described in the caption, clearly mapping raw inputs, intents, tasks, and structured outputs across four distinct domains. The layout is uncluttered, and the visual elements (video thumbnails, text blocks) are legible and well-aligned with the textual descriptions.

## Figure 8

The figure provides a clear and comprehensive visual overview of the DataClaw pipeline, but the provided caption text is incomplete, cutting off mid-sentence at the end.

## Figure 9

The figure effectively displays data distribution and scaling curves, but the caption is poorly edited with repetitions and typos. Additionally, the t-SNE visualization omits the ‘DataClaw-O’ model mentioned in the caption, creating a discrepancy between the text and the visual evidence.

## Figure 10

The figure effectively visualizes model performance comparisons with clear annotations, but the caption inaccurately describes the panels as ‘construction’ and ‘reconstruction’ processes rather than qualitative evaluations. Additionally, the critique for one model output contradicts the visible JSON content.

**Action items.**

- **[writing]** Figure 1: The caption describes the figure as a ‘Qualitative example for GUI navigation,’ but the image displays a grid of 32 distinct steps with code snippets. The caption is too brief to explain the grid layout or the specific task (Home Depot truck rental) shown in the screenshots.
- **[writing]** Figure 1: The text within the image (e.g., ‘pyautogui.click(x=0.9491, y=0.2208)’) is extremely small and illegible in the rendered view, making it difficult to verify the specific coordinates or actions described.

- **[science]** Figure 2: The caption claims a ‘Qualitative comparison for action video generation,’ but the figure displays static frames from a robot manipulation task (pouring liquid) rather than generated video sequences or a comparison of generation quality.
- **[writing]** Figure 2: The row labels (e.g., ‘Qwen3.5-9b’, ‘Gemini-3.1-pro-preview’) identify specific models, but the caption fails to define what these rows represent (e.g., ‘Generated by’, ‘Baseline’, ‘Ground Truth’), leaving the viewer to guess the role of each row.
- **[science]** Figure 3: The image displays a multiple-choice question with a provided answer string, but lacks the specific ‘Question’ and ‘Answer’ labels or structured layout described in the main Figure 7 caption format for qualitative examples, making it unclear if this is a raw prompt or a processed result.
- **[writing]** Figure 3: The text block is dense and contains raw newline characters (e.g., ‘\nPlease select...’) that should be formatted as a list or paragraph for readability.
- **[science]** Figure 4: The ‘model response’ JSON claims the input video is frames [2,9] and output is [10,18], but the visual evidence shows the character running forward in the input frames (2, 9) and running backward in the output frames (10, 18). The Chain-of-Thought reasoning (‘running forward... crossing the boulder slope’) contradicts the visual content of the output frames, indicating a failure in the ‘tailoring’ or a hallucination in the reasoning process.
- **[writing]** Figure 4: The caption describes a ‘Successful world-model or video-generation tailoring case’, but the figure displays a text-based JSON response rather than a generated video or image sequence, making it unclear how the ‘video-generation’ claim is visually supported.
- **[science]** Figure 5: The ‘model response’ JSON contains ‘output\_image’: [10,17,23], but the answer text explicitly describes ‘Step 1’ (sprints) and ‘Step 2’ (takes off) which correspond to frames 0 and 5, yet these are missing from the output list and the answer text does not reference them.
- **[writing]** Figure 5: The ‘model response’ answer text includes raw HTML-style tags ‘<img>’ instead of referencing the specific frame numbers or images, which reduces clarity and contradicts the structured JSON format.
- **[science]** Figure 6: The caption describes a ‘Failed daily-life tailoring case,’ but the model response JSON contains a valid, coherent Chain-of-Thought and navigation instruction. The output does not demonstrate a failure (e.g., hallucination, missing data, or incorrect reasoning) as implied by the figure title.
- **[writing]** Figure 6: The ‘input\_image’ array in the JSON response lists indices [0,1,2...20], but the visual ‘user input’ section only displays 8 specific frames (0, 6, 10, 11, 15, 19, 21, 25). The discrepancy between the listed indices and the shown frames is unexplained.
- **[writing]** Figure 8: The caption text ends abruptly mid-sentence (‘...routed’) and appears to be missing the final clause describing the routing mechanism.

- **[writing]** Figure 9: The caption contains a stray ‘%’ symbol after the first sentence and repeats the description of the right panels twice.
- **[science]** Figure 9: The t-SNE visualization legend lists ‘Raw Data’, ‘DataClaw-E’, and ‘Qwen3.5-9B’, but the caption claims to compare ‘DataClaw-O’ as well, which is missing from the plot.
- **[science]** Figure 9: The t-SNE plot lacks a clear title or axis labels indicating what specific model representations or data subsets are being visualized, making the comparison ambiguous.
- **[writing]** Figure 10: The caption states the left panel shows ‘robot manipulation data construction,’ but the image displays a comparison of model outputs (Gemini, Qwen, DataClaw) for a specific task, which is a qualitative evaluation rather than a construction process visualization.
- **[writing]** Figure 10: The caption states the right panel shows ‘GUI task reconstruction,’ but the image shows a comparison of model outputs for a task ordering problem, which is a qualitative evaluation rather than a reconstruction process visualization.
- **[science]** Figure 10: The red cross annotation for Qwen3.5-9b on the left panel claims ‘The constructed data does not include questions or answer,’ yet the JSON output shown explicitly contains ‘question’ and ‘answer’ fields, creating a contradiction between the critique and the visual evidence.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The manuscript relies heavily on undefined jargon and proprietary-sounding terminology that creates a barrier for non-specialist readers. In the Introduction, the term “data entropy” is used to describe noisy streams but is never defined; “high noise and redundancy” would be clearer. Similarly, “Factual Anchors” is introduced as a capitalized concept without a plain-language definition. The core contribution, “Agentic Data Tailoring,” is presented as a proper noun without explaining the underlying mechanism in accessible terms.

Throughout the Method and Experiments sections, acronyms are frequently used without expansion. “GRPO” (Group Relative Policy Optimization) appears in the Introduction and Section 3.3 without being spelled out. “Omni” and “Expert” are used as model names (DataClaw0-O/E) rather than descriptive terms like “unified” and “domain-specific.” In the evaluation sections, metrics like “SSR” and “TSR” are used in tables and text without being defined at first use. Additionally, “CoT” is used for “chain-of-thought” without expansion, and “AST” (Abstract Syntax Tree) appears in the appendix without definition.

The term “Fuzzy-intent” is used as a specific benchmark category without a clear definition

for what constitutes a “fuzzy” intent versus a standard one. Finally, phrases like “temporal shape similarity” and “trajectory-shape similarity” are overly technical; “temporal alignment” or “sequence matching” would be more accessible. These terms must be defined or replaced to ensure the paper is understandable to a broader audience.

### Action items.

- **[writing]** Define ‘data entropy’ in the Introduction (Section 1) or replace with ‘high noise and redundancy’ to avoid undefined jargon.
- **[writing]** Define ‘Factual Anchors’ upon first use in Section 1 or Section 3.1; currently introduced as a capitalized concept without explanation.
- **[writing]** Replace ‘Agentic Data Tailoring’ with ‘agent-driven data refinement’ or define the term explicitly in the Introduction to aid non-specialist readers.
- **[writing]** Define ‘GRPO’ (Group Relative Policy Optimization) at first mention in Section 1 or 3.3; currently used as an acronym without expansion.
- **[writing]** Replace ‘Omni’ and ‘Expert’ paradigms with ‘unified’ and ‘domain-specific’ models in Section 1 and 3.5 to reduce proprietary-sounding jargon.
- **[writing]** Define ‘Fuzzy-intent’ in Section 1 or Appendix B.1; the term is used as a specific category without a clear definition for general readers.
- **[writing]** Replace ‘temporal shape similarity’ and ‘trajectory-shape similarity’ with ‘temporal alignment’ or ‘sequence matching’ in Section 4.3 and Appendix B.
- **[writing]** Define ‘SSR’ and ‘TSR’ (Step/Task Success Rate) at first use in Section 4.1 or Appendix D.1; currently used as acronyms without expansion.
- **[writing]** Replace ‘CoT’ with ‘chain-of-thought reasoning’ at first use in Section 3.3 or Appendix B to ensure clarity for non-LLM specialists.
- **[writing]** Define ‘AST’ (Abstract Syntax Tree) in Appendix B.1 when discussing format compliance; the acronym is used without definition.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical consistency of the paper is generally sound, with the proposed “Agentic Data Tailoring” framework following a coherent path from problem formulation to solution and evaluation. The two-stage pipeline (Factual Anchors + Semantic Synthesis) logically supports the claim of reducing data entropy. The use of GRPO with deterministic rule-based rewards is a consistent methodological choice that aligns with the goal of schema compliance.

However, there are minor inconsistencies between the textual claims and the presented data in the tables that require clarification:

1. **Claim vs. Data in Main Results:** In Section 5.2, the text states, “It leads in Sequence scores for Embodied (71.60) and Fuzzy (50.31).” While DataClaw0-E (71.60) is indeed higher than Gemini (67.97) and GPT-4o (46.33), the phrasing “leads in” without qualification could be misinterpreted as a universal lead over all proprietary models if the reader does not cross-reference the full table. The text should explicitly state “leads among the evaluated open-source models” or “surpasses the strongest proprietary baseline in these specific categories” to be logically precise.
2. **Ablation Interpretation:** In Section 5.5, the text claims that wrong routing causes “severe degradation (e.g., GUI expert on Embodied task: 0.00 Field).” Table 3(b) confirms the 0.00 Field score for “Embodied (Gui)” but shows a Sequence score of 50.00. The text implies a total collapse, whereas the data shows a partial failure (schema adherence failed, but sequence generation was non-zero). The conclusion should be refined to reflect that routing errors specifically destroy schema validity (Field score) rather than all generative capabilities.
3. **Scaling Law Causality:** The assertion that “task interference” causes the oscillation in DataClaw0-O (Section 5.4) is a plausible hypothesis, but the paper does not provide a mechanistic explanation (e.g., gradient conflict analysis or attention map divergence) in the main text to strictly support this causal claim. The evidence is correlational (oscillation exists) rather than causal (interference causes oscillation). While acceptable for a high-level summary, a brief mention of the mechanism would strengthen the logical link.

These issues are primarily matters of precise phrasing and the strength of causal assertions rather than fundamental logical flaws. The core argument remains valid.

#### Action items.

- **[writing]** The claim that DataClaw0-E leads in Sequence scores for Embodied (71.60) and Fuzzy (50.31) contradicts Table 1, where Gemini-3.1-Pro-Preview scores 67.97 and 47.25 respectively. While DataClaw0-E is higher, the text implies a lead over all baselines, but GPT-4o scores 46.33 (Embodied) and 42.57 (Fuzzy). The phrasing needs precise qualification to avoid overclaiming against specific baselines not listed in the immediate context.
- **[writing]** The ablation study in Table 3(b) claims expert routing is indispensable, citing a drop to 0.00 Field score when routing is wrong. However, the table shows ‘Embodied (Gui)’ yields 0.00 Field but 50.00 Sequence. The conclusion that routing causes ‘severe degradation’ is supported, but the text implies total failure across all metrics, which is not strictly true for the Sequence metric in that specific row.
- **[writing]** The paper states DataClaw0-O shows ‘unstable scaling’ due to task interference, citing scores 53.60 -> 47.23 -> 57.84. However, the text does not explicitly define the x-axis (data scale) for these points in the main text, relying on the reader to infer from the Appendix or Figure 4. The causal link between ‘task interference’ and the specific oscillation pattern is asserted but not mechanistically explained in the main text.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper exhibits significant overreach in its comparative claims and the characterization of its contributions.

First, the central narrative that DataClaw0 “matches” or “outperforms” proprietary state-of-the-art models (specifically Gemini-3.1-Pro and GPT-4o) is not fully supported by the aggregate data. In Section 4.2 and Table 2, the authors highlight that DataClaw0-E achieves superior results on specific end-to-end metrics (e.g., GUI TSR, VQA Overall Accuracy). However, the same table explicitly shows that DataClaw0-E underperforms Gemini-3.1-Pro on critical step-level metrics (GUI SSR: 38.2% vs 39.5%) and partial accuracy (VQA Partial Acc: 52.1% vs 53.4%). The conclusion that the method “matches” or “outperforms” proprietary annotators is a selective interpretation of the results that ignores these deficits. The claim should be tempered to reflect a trade-off (e.g., “competitive on end-to-end success but lagging on step-level precision”) rather than a blanket statement of superiority.

Second, the introduction of “DataClaw0-val” as the “first benchmark for data refinement” (Introduction) is an overreach given the evaluation methodology. While the benchmark covers five domains, the evaluation for the “Fuzzy” subset relies on a user study with only 100 participants (Appendix e001). The paper presents these results as definitive evidence of robustness to fuzzy intents without providing statistical significance testing, inter-annotator agreement metrics, or a discussion on the variance of human judgments. Claiming a new benchmark is a standard for the field based on a small-scale, non-standardized human evaluation is premature.

Finally, the claim of “efficiency” and “reduced costs” (Introduction, Conclusion) is unsupported. The proposed pipeline involves a two-stage process: extracting anchors with lightweight experts and then synthesizing data with strong VLMs, followed by GRPO training. The paper does not provide a cost analysis comparing the compute and API costs of this multi-step agentic pipeline against the direct usage of the proprietary models (Gemini/GPT-4o) used as baselines. Without this comparison, the assertion that the method offers a “scalable, high-value paradigm” with “reduced costs” is an extrapolation beyond the provided evidence.

### Action items.

- **[science]** The claim that DataClaw0 ‘matches proprietary annotators’ (Conclusion) and ‘outperforms’ them on end-to-end metrics (Section 4.2) is unsupported. Table 2 shows DataClaw0-E trailing Gemini-3.1-Pro on GUI SSR (38.2% vs 39.5%) and VQA Partial Accuracy (52.1% vs 53.4%). The paper selectively highlights only the metrics where it wins (TSR, FVD) to construct a narrative of superiority that the full data table contradicts.
- **[science]** The benchmark ‘DataClaw0-val’ is defined as the ‘first benchmark for data refinement’ (Introduction), yet the evaluation relies heavily on a 200-sample set with fuzzy-intent stress tests evaluated via a 100-user user study (Appendix e001). The paper overstates the

rigor of this benchmark by presenting it as a definitive standard without addressing the statistical power or reproducibility of the human evaluation component.

- **[writing]** The paper claims to demonstrate ‘Targeted Refinement’ that distills ‘superior training value’ (Introduction), but the downstream experiments (Table 3) use a ‘strict volume alignment’ protocol that is not fully justified. The claim that the method is ‘cost-effective’ is an overreach because the computational cost of the two-stage pipeline (lightweight experts + strong VLM synthesis + GRPO) is not compared against the cost of the proprietary baselines (Gemini/GPT-4o) used for comparison.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript proposes “Agentic Data Tailoring” to synthesize structured multimodal data from raw streams. While the technical approach is novel, the **safety and ethics** section requires specific clarifications regarding data provenance, human subject involvement, and potential dual-use risks.

**1. Human Subject Research and Consent** In Appendix e001 (Section “DataClaw0-Intent: Fuzzy-intent Stress Test”), the authors state: “the final evaluation is conducted through user study. 100 human users judge whether the model’s constructed sample is useful...”

- **Concern:** There is no mention of Institutional Review Board (IRB) approval, ethical clearance, or informed consent procedures for these 100 participants.
- **Requirement:** The authors must add a statement confirming whether this study was exempt from IRB review or if formal approval was obtained. Additionally, details on how informed consent was obtained and how participant data was anonymized must be included.

**2. Data Privacy and Provenance** The paper claims to process “raw streams” from diverse domains including “Daily Life” (e001) and “Education” (e001).

- **Concern:** The source of these raw streams is not explicitly defined as public, anonymized, or synthetic. If the raw streams contain real-world video of individuals (e.g., from Ego4D or similar sources), there is a risk of re-identification or privacy leakage if the “tailoring” process inadvertently preserves sensitive metadata or visual details.
- **Requirement:** The authors must explicitly state the privacy safeguards applied to the raw data. Did they perform face blurring, audio redaction, or metadata stripping? A statement confirming that the dataset complies with relevant privacy regulations (e.g., GDPR, CCPA) or that the data is strictly public domain is necessary.

**3. Safety in Embodied AI Data Synthesis** The “Robot Data Pipeline” (Appendix e002)

describes a “Fault synthesis branch” where the system “Simulates stalls by repeating frames” to create fault-diagnosis data.

- **Concern:** While this appears to be a data augmentation technique, the paper does not clarify if the underlying raw data was collected from real robots operating in physical environments. If real robots were used to generate the “stall” data, there is a potential for physical harm or equipment damage if the data collection protocol was not strictly controlled.
- **Requirement:** Clarify the origin of the robot data. If it comes from public datasets (e.g., RoboCoin, Agibot), cite the specific dataset and its safety protocols. If the authors collected new data, confirm that no physical harm occurred to robots or humans during the “fault” simulation.

**4. Dual-Use and Misinformation Risks** The system is designed to generate “structured supervision” and “video outputs” (Figure 1, e000) that can reconstruct or synthesize video sequences.

- **Concern:** The ability to generate high-quality, schema-aligned video-text pairs from raw streams could be misused to create synthetic training data for deepfake generation or to automate the creation of misleading instructional content (e.g., fake repair tutorials).
- **Requirement:** While not a reason for rejection, the authors should include a brief “Limitations and Risks” paragraph discussing the potential for their “Agentic Data Tailoring” framework to be repurposed for generating synthetic media that could be used for misinformation or disinformation campaigns.

The paper is otherwise sound in its technical claims, but these ethical disclosures are mandatory for publication in a venue concerned with responsible AI.

#### Action items.

- **[writing]** The paper lacks explicit IRB/ethics approval statements for the human user study (100 users) used to evaluate the ‘DataClaw0-Intent’ benchmark (Appendix e001). Authors must confirm IRB exemption or approval and detail informed consent procedures.
- **[writing]** The ‘Robot Data Pipeline’ (Appendix e002) describes synthesizing ‘fault’ scenarios by repeating frames to simulate stalls. Authors must clarify the source of the raw robot data (e.g., public datasets vs. private recordings) and confirm that no real-world robots were put at risk or damaged during the data collection or synthesis process.
- **[writing]** The paper mentions processing ‘raw streams’ from domains like ‘Daily Life’ and ‘Education’ without specifying data privacy protocols. Authors must explicitly state whether personally identifiable information (PII) or private user data was redacted or excluded from the training corpus to prevent privacy violations.

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The scientific evidence supporting the central claims of DataClaw0 is currently insufficient to justify the strong conclusions drawn regarding its superiority over proprietary baselines and its generalizability.

First, the statistical rigor of the downstream evaluation (Table 2, Section 5.3) is weak. The paper claims DataClaw<sub>0</sub> outperforms Gemini-3.1-Pro on end-to-end metrics (e.g., VQA Overall Accuracy: 33.2% vs 31.5%; GUI TSR: 15.6% vs 14.2%). These differences are marginal (1-2 percentage points). The manuscript fails to report standard deviations, confidence intervals, or results from statistical significance tests (e.g., paired t-tests) across multiple random seeds. Without this, it is impossible to determine if these gains are real or artifacts of random initialization or specific data splits. The claim of “superiority” is not statistically supported.

Second, the primary benchmark, DataClaw<sub>0</sub>-val, relies on a sample size of only 200 examples (Appendix e001). This is statistically underpowered for evaluating performance across five distinct and complex domains (GUI, Embodied, AIGC, Daily, Education). With such a small N, the reported scores (e.g., 97.53% Field score) are highly susceptible to variance and potential overfitting to the specific test cases. The authors must either significantly expand the benchmark to a statistically robust size (e.g., >1000 examples) or provide a rigorous power analysis demonstrating that 200 samples are sufficient to detect the reported effect sizes with high confidence.

Third, the evaluation of the “Fuzzy” intent capability relies entirely on a user study with 100 participants (Appendix e001). The methodology lacks critical details: there is no report of inter-annotator agreement (e.g., Cohen’s Kappa) to establish the reliability of the human judgments, nor is the scoring rubric explicitly defined. This introduces significant subjectivity, making the claim that the model “rivals Gemini” on fuzzy intents scientifically unverifiable.

Finally, the ablation studies (Table 3) demonstrate the necessity of components but lack depth in quantifying robustness. For instance, the routing ablation shows a 0.00 score for wrong routing, which is expected, but does not quantify the performance degradation under realistic, non-catastrophic routing errors or compare against a random baseline to isolate the routing module’s specific contribution.

To proceed, the authors must re-run downstream experiments with multiple seeds to report variance and statistical significance, expand the benchmark size, and provide rigorous methodological details for the user study.

### Action items.

- **[science]** The scientific evidence supporting the central claims of DataClaw0 is currently insufficient to justify the strong conclusions drawn regarding its superiority over proprietary baselines and its generalizability. First, the statistical rigor of the downstream evaluation

(Table 2, Section 5.3) is weak. The paper claims DataClaw<sub>0</sub> outperforms Gemini-3.1-Pro on end-to-end metrics (e.g., VQA Overall Accuracy: 33.2% vs 31.5%; GUI TSR: 15.6% vs 14.2%). These differences are marginal (1-2 perce

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical analysis in this manuscript is currently insufficient to support the strong claims of superiority and robustness. While the experimental design is ambitious, the reporting of results lacks the necessary statistical depth required for a rigorous scientific evaluation.

First, the primary benchmark results presented in Table 1 (Main Results) are derived from a single deterministic pass of the models. In the context of Large Language Model (LLM) and Vision-Language Model (VLM) generation, outputs are inherently stochastic. Reporting a single point estimate without confidence intervals, standard deviations, or results averaged over multiple random seeds makes it impossible to assess the reliability of the reported scores. For instance, the difference between DataClaw0-E (74.94) and GPT-4o (75.15) in the Semantic metric is negligible and likely within the margin of error for a single run. The authors must re-run evaluations with multiple seeds (e.g., 5-10) and report the mean and standard deviation, or provide bootstrapped confidence intervals, to substantiate claims of “competitive” or “superior” performance.

Second, the evaluation of the DataClaw0-Intent benchmark (Appendix) relies entirely on a user study with 100 participants. The manuscript states that users judged the “usefulness” of outputs but fails to report any measure of inter-annotator agreement (e.g., Cohen’s Kappa or Fleiss’ Kappa). Without this, the reliability of the human labels is unknown. Furthermore, the claim that DataClaw0 “significantly surpasses” the base model is unsupported by any statistical hypothesis testing (e.g., McNemar’s test or a t-test on the user ratings). The absence of p-values or effect sizes renders this comparison anecdotal rather than statistical.

Third, the downstream application results in Table 2 show marginal improvements (e.g., GUI TSR 15.6% vs 14.2%). These differences are small and could easily be attributed to the variance inherent in the downstream fine-tuning process (random weight initialization, batch ordering, etc.). The authors do not report the variance across multiple fine-tuning runs. To claim that DataClaw0-generated data is “superior,” the authors must demonstrate that these gains are statistically significant, likely requiring multiple independent training runs for each data source.

Finally, the ablation studies in Table 3 (Reward Design and Expert Routing) present single-point results. The conclusion that the anchor reward is “critical” based on a Sequence score increase from 70.11 to 71.96 is weak without error bars. Given the sensitivity of Reinforcement Learning (GRPO) to hyperparameters and initialization, a single run is not representative. The authors should report the variance of these ablation results to ensure the observed improvements

are robust.

In summary, the paper requires a full revision of its experimental section to include proper statistical reporting: multiple seeds for all automated evaluations, confidence intervals, inter-annotator agreement for human studies, and significance testing for all comparative claims.

#### Action items.

- **[science]** The statistical analysis in this manuscript is currently insufficient to support the strong claims of superiority and robustness. While the experimental design is ambitious, the reporting of results lacks the necessary statistical depth required for a rigorous scientific evaluation. First, the primary benchmark results presented in Table 1 (Main Results) are derived from a single deterministic pass of the models. In the context of Large Language Model (LLM) and Vision-Language Model (VLM) genera

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript presents a novel framework for agentic data tailoring, but the writing quality is currently compromised by structural redundancies and inconsistent terminology that hinder readability.

The most critical issue is the presence of duplicate sections. The Introduction, Related Work, and Method sections appear to be pasted twice in the source (evident in the transition from chunk ‘e000’ to ‘e003’), resulting in repeated definitions of “Agentic Data Tailoring” and the “DataClaw0” pipeline. This duplication breaks the narrative flow and forces the reader to sift through redundant information. The authors must consolidate these sections into a single, linear progression.

Additionally, there are inconsistencies in model naming and variable definitions. In Table 1 and the accompanying text, the baseline “GPT-4o (1120-global)” is introduced without a clear explanation of the “(1120-global)” suffix in the main text, which may confuse readers regarding the experimental setup. While the table caption mentions “state-of-the-art MLLM models,” the specific configuration of this baseline should be explicitly defined in the “Experimental Setup” subsection to ensure clarity.

Finally, the mathematical exposition in the “Rule-Driven Reinforcement Learning” section is dense. While the equations are correct, the surrounding prose often jumps between the definition of the reward function, the advantage estimation, and the optimization objective without sufficient transitional sentences. Breaking these complex derivations into smaller, step-by-step explanations would improve the accessibility of the method for a broader audience.

Overall, the scientific content is promising, but the manuscript requires a thorough pass to remove duplicates, standardize terminology, and smooth out the technical exposition.

#### Action items.

- **[writing]** The manuscript contains duplicate content where the Introduction, Related Work, and Method sections appear twice (e.g., ‘e000’ vs ‘e003’). This creates significant confusion and disrupts the logical flow. The authors must consolidate these sections into a single, coherent narrative.
- **[writing]** In the ‘Experiments’ section, the text references ‘GPT-4o (1120-global)’ in Table 1 and the main text, but the model name is inconsistent with standard naming conventions and lacks a clear definition of what ‘1120-global’ signifies in the context of the benchmark setup.
- **[writing]** Several sentences in the ‘Method’ and ‘Appendix’ sections are overly dense with mathematical notation and acronyms (e.g., the GRPO objective function description) without sufficient transitional prose to guide the reader through the derivation. Consider breaking these into smaller, more digestible steps.